DA3452 Speech and Language Technology

Module 5 Test

BERT, prompting, scaling and supervised adaptation through LoRA

Duration 60 minutes Maximum marks 60 Questions 31

Name:	Roll No.:	Date:

Instructions

- Section A: 15 MCQs, 25 marks. Section B: 7 MSQs, 14 marks. Section C: 9 fill-in-the-blank questions, 21 marks. All questions are compulsory; marks are shown against each question.
- MCQ: Select one option.
- MSQ: Select all correct options. An MSQ may have one or more correct options.
- MSQ marking: Selecting any incorrect option gives 0 for that question. Otherwise, marks are awarded in proportion to the correct options selected (rounded to the nearest 0.5). There is no negative marking.
- Record final answers on the response sheet. Token positions are numbered from 1 unless a question states otherwise.
- Use decimal GB = 10^9 bytes where memory is requested. Use the stated LoRA row-vector convention, with no scaling factor unless specified.
- **As announced, malpractice can result in disciplinary actions as per the Senate mandate.**

SECTION A MCQ

Questions 1-15 | 25 marks

1. [1 mark] A team replaces the original BERT pretraining objectives with the RoBERTa objective configuration discussed in class. Which change is appropriate?

A. Remove MLM and retain next-sentence prediction.

B. Replace bidirectional attention with causal attention.

C. Retain next-sentence prediction and replace MLM with instruction tuning.

D. Retain MLM and remove next-sentence prediction.

2. [1 mark] System E classifies a review using an encoder and a three-class task head. System G generates a sentiment label using a decoder-only language model whose vocabulary has 32,000 tokens. Before any restriction to allowed labels, how many logits does each head produce at one prediction step? Give (E, G).

A. (3, 3)

B. (3, 32,000)

C. (32,000, 3)

D. (32,000, 32,000)

3. [**1 mark**] A prompt contains:

```
Label the ticket as Billing, Technical or Other.
Ticket: I was charged twice.
Label: Billing
Ticket: The application will not open.
Label: Technical
Ticket: Please change my postal address.
Label:
```

How many completed demonstrations does this prompt contain?

A. Zero

B. One

C. Two

D. Three

4. [**2 marks**] The table shows five positions from a longer original-BERT training sequence. Positions 2 and 4 are selected for MLM prediction. Position 2 receives a random replacement; position 4 is left unchanged.

Position	1	2	3	4	5
Original token	The	cat	slept	near	home
Encoder input	The	dog	slept	near	home

Which **(prediction positions; target tokens)** pair belongs to the MLM loss?

A. (2, 4; cat, near)

B. (2; cat)

C. (2, 4; dog, near)

D. (1, 2, 3, 4, 5; The, cat, slept, near, home)

5. [**1 mark**] Two copies, P and F, start from the same language-model checkpoint. P answers requests using solved demonstrations in its prompt. F is supervised fine-tuned on the same examples and makes a nonzero parameter update. Each system then saves its model parameters; prompts, activations and caches are discarded. Which saved parameter file is identical to the starting checkpoint?

A. F's file only.

B. Both saved files.

C. Neither saved file.

D. P's file only.

6. [**2 marks**] A researcher evaluates 30 prompt variants on every example in a labelled test set T and chooses the highest-scoring prompt. The prompt is now frozen. Which next action provides a final evaluation that was not used to select it?

A. Evaluate it on T with a different random decoding seed.

B. Evaluate it on fresh test inputs from the target population.

C. Evaluate it on a randomly chosen half of the same T.

D. Evaluate it on T after randomly rearranging its input order.

7. [**2 marks**] A teacher receives a user instruction x together with a long rubric R and produces response y_R. Without R, it produces response y_0. A student should learn the rubric-guided behaviour while receiving only x at deployment. Which student training record implements this context-distillation plan?

A. Input: x; target: y_R.

B. Input: R followed by x; target: y_R.

C. Input: x; target: y_0.

D. Input: R; target: y_R.

8. [**2 marks**] An instruction-tuning collection contains four textual-entailment datasets, three sentiment datasets and two summarisation datasets. The aim is to evaluate generalisation to an unseen task type, specifically textual entailment. Which split best matches that aim?

A. Withhold 20% of the examples from each entailment dataset.

B. Withhold one entailment dataset; train on the other three.

C. Withhold all entailment datasets; train on sentiment and summarisation.

D. Withhold one instruction template; retain examples from every task type.

9. [**2 marks**] Two sentiment prompts are evaluated on the same 100 reviews. The fixed scoring rule accepts only an exact label; an invalid output counts as incorrect.

Prompt	Valid label outputs	Correct label outputs
P	80	70
Q	95	70

A report drops invalid outputs and quotes accuracies of 70/80 for P and 70/95 for Q. Under the declared scoring rule, what are the task accuracies? Give (P, Q).

A. (87.5%, 73.7%)

B. (70%, 70%)

C. (87.5%, 70%)

D. (70%, 73.7%)

10. [**2 marks**] Two inputs have identical token IDs through position i, including the token at i, but differ after i. Process each input using (E) the same bidirectional BERT encoder and (D) the same causal decoder. Position IDs are matched; computation is deterministic, with no padding. Which statement about the final hidden state at i is warranted?

A. It must be identical in E and must differ in D.

B. It must differ in both E and D.

C. It can differ in E; it is identical in D.

D. It must be identical in both E and D.

11. [2 marks] Three dense-model runs have the same estimated training-compute budget. Their measured losses use the same held-out text and tokenizer.

Run	Parameters N	Training-token presentations D	Mean loss
P	2.5 billion	80 billion	2.20
Q	5 billion	40 billion	2.10
R	10 billion	20 billion	2.00

Deployment requires mean loss **no greater than 2.15**. All systems will process the same total T inference tokens. Using Cinfer ≈ 2NT, which run meets the quality requirement with the lowest estimated inference compute?

A. Run P.

B. Run Q.

C. Run R.

D. No run meets the requirement.

12. [2 marks] A PEFT system stores 10^9 frozen base parameters and 10^7 trainable adapter parameters. Every stored value is FP32, using 4 bytes. Frozen parameters require weights only; each trainable parameter requires a weight, gradient and two Adam moments. Excluding activations and overheads, how much memory is required? Use decimal GB.

A. 0.16 GB

B. 4.04 GB

C. 4.16 GB

D. 16.16 GB

13. [1 mark] System E has a trained three-class sentiment head for Positive, Negative and Neutral. System G is a decoder-only language model that generates these labels through its existing vocabulary head. A revised task adds the label Mixed, whose text can already be encoded with G's existing vocabulary. Which change lets each system represent the four-label task without changing G's architecture?

A. E keeps its three outputs; G adds Mixed to its prompt.

B. E expands to four outputs; G expands its vocabulary head to four outputs.

C. E keeps its three outputs; G adds a new token embedding for Mixed.

D. E expands and retrains its class head; G adds Mixed to the prompt.

14. [2 marks] A trained LoRA layer uses h = xW_0 + xAB, with the factors below. W_0 is the frozen base weight.

$$A=\begin{bmatrix}1\\-1\end{bmatrix},\quad B=\begin{bmatrix}1 & 2\end{bmatrix}$$

For which input does the adapter leave the base-layer output unchanged?

A. x = [1, 0]

B. x = [0, 1]

C. x = [1, 1]

D. x = [1, −1]

15. [**2 marks**] A decoder-only language model is being fine-tuned with LoRA adapters in several layers. The base weights remain frozen. The input x to the current adapted layer comes from earlier layers that contain trainable LoRA adapters. Two implementations use the same W_0, A, B, x and loss.

- **F:** Keep W_0 fixed and backpropagate through both branches of $h = xW_0 + xAB$.
- **D:** Keep W_0 fixed, but treat xW_0 as a constant during backpropagation.

For this forward-and-backward pass, before any parameter update, which quantity can differ between F and D?

A. The forward output h.

B. The gradient $\partial L/\partial A$.

C. The gradient $\partial L/\partial B$.

D. The gradient $\partial L/\partial x$.

SECTION B MSQ

Questions 16-22 | 14 marks

16. [**2 marks**] Original BERT selects some token positions for MLM prediction. A selected position may be replaced by [MASK], replaced by a random token, or left unchanged. Which statements are correct?

A. A selected position left unchanged still contributes to MLM prediction loss.

B. At a selected position replaced by a random token, the target becomes that random replacement.

C. BERT can use tokens on both sides of a selected position when computing its contextual representation.

D. An unselected position contributes its own token-prediction term to MLM loss simply because it is visible to the encoder.

17. [**2 marks**] Two copies of the same pretrained decoder-only language model use its existing vocabulary head. System P receives two solved examples in its prompt and performs inference with fixed weights. System F is supervised fine-tuned on the same instruction–response examples. Which statements are correct?

A. Saving P's unchanged checkpoint stores the demonstrations for use in future prompts.

B. Adding demonstrations to P can change its intermediate hidden states and predictions.

C. Training F requires changing its vocabulary size to the number of task labels.

D. Saving F after gradient updates retains the parameter changes learned from the examples.

18. [**2 marks**] Two instruction wordings are evaluated using the same model version, decoding settings and labelled inputs. Which choices support a sound comparison?

A. Retain the two predictions for each input so that the comparison can account for paired outcomes.

B. Repeatedly revise instructions after inspecting final-test answers, then report the best final-test accuracy as an unbiased estimate.

C. Specify the parsing and scoring rule before examining comparison results.

D. Keep solved demonstrations separate from the inputs used to evaluate performance.

19. [**2 marks**] Use the rough dense-model estimate $C \approx 6ND$, where N is parameter count and D counts training-token presentations. Which statements are correct?

A. Two allocations with the same ND must achieve the same final loss.

B. At fixed estimated compute, doubling N halves the affordable D.

C. Processing the same corpus for two complete passes doubles D relative to one pass.

D. The expression directly gives wall-clock training time, independent of hardware.

20. [**2 marks**] A team constructs an instruction-tuning set from labelled NLP records and teacher-generated responses. Which statements are correct?

A. Converted NLP records can use their existing ground-truth labels as target responses.

B. Filtering targets for valid output format also removes the teacher's factual errors.

C. A student trained on unchecked teacher errors can learn to reproduce those errors.

D. Rare cases can disappear when generated samples replace the training data repeatedly.

21. [**2 marks**] A LoRA update is $\Delta W = AB$, with A: 3×1 and B: 1×3. Which statements are correct? Here diag(a, b, c) denotes a diagonal matrix with those three entries.

A. $\Delta W = \text{diag}(1, 0, 0)$ can be represented exactly.

B. $\Delta W = \text{diag}(1, 1, 0)$ can be represented exactly.

C. Each column of ΔW is proportional to the column vector A.

D. At most one column of ΔW can contain nonzero entries.

22. [**2 marks**] Two LoRA adapters, U and V, were trained separately from the same base weight W_0. An engineer merges U into W_0. They retain the original base checkpoint and both adapters. They now want the effective weight for V trained from the original base.

Assume W_0 and U's update are nonzero. Ignore numerical rounding. Which procedures recover the intended effective weight?

A. Reload W_0, then activate V's adapter.

B. Subtract U's update from the merged weight, then activate V's adapter.

C. Keep U's merged weight, then activate V's adapter.

D. Discard the merged weight and use V's update as the complete weight.

SECTION C FILL IN THE BLANKS

Questions 23-31 | 21 marks

23. [**2 marks**] A service processes 10^6 requests. Each request contains 900 prompt tokens and generates 100 response tokens. For the estimate $C_{infer} \approx 2NT$, the processed-token count T is _____ **tokens**. Counting only generated tokens would underestimate this compute by a factor of _____.

24. [**2 marks**] A dense model has 2×10^9 parameters. Training makes three complete passes through a corpus containing 10^{10} tokens. In $C \approx 6ND$, D = _____ **token presentations** and $C \approx$ _____ **FLOPs**.

25. [**2 marks**] Use both approximations $C \approx 6ND$ and $D \approx 20N$. For $C = 1.2 \times 10^{20}$ FLOPs, the corresponding allocation is approximately _____ **parameters** and _____ **training-token presentations**.

26. [**2 marks**] Assume full-FP32 Adam training. A model has 2×10^9 trainable parameters. What is the combined storage for its weights, gradients and two Adam moments?

Combined storage = _____ **GB**.

Give the numerical value in GB, using 1 GB = 10^9 bytes. Exclude activations and overheads.

27. [**2 marks**] Use $C_{train} \approx 6ND$ and $C_{infer} \approx 2NT$, where T counts all processed inference tokens. A model was pretrained on $D = 5 \times 10^{10}$ token presentations. Its estimated cumulative inference compute equals its estimated pretraining compute when T = _____ **tokens**.

28. **[2 marks]** An instruction-SFT example contains four prompt tokens, two response tokens and three padding tokens. The correct response-token probabilities under teacher forcing are 0.5 and 0.25. Use a **summed response-only loss**. Given ln 2 = 0.693, the loss is _____ **nats**.

29. **[2 marks]** LoRA is applied to all four attention projection matrices in one block. Each matrix is 512 × 512 and has its own factors A: 512 × 4 and B: 4 × 512. The total number of trainable factor entries across the four adapted matrices is _____.

30. **[2 marks]** Use row vectors and $h = xW_0 + xAB$. W_0 is frozen. The adapter factors and upstream gradient are:

$$x=\begin{bmatrix}1 & 1\end{bmatrix},\quad A=\begin{bmatrix}1\\-1\end{bmatrix},\quad B=\begin{bmatrix}1 & 2\end{bmatrix},\quad g=\begin{bmatrix}1 & -1\end{bmatrix}$$

Here $g = \partial L/\partial h$ and $xA = 0$, so the adapter's current forward contribution is zero. In backpropagation:

The first entry of $\partial L/\partial A$ is _____.

The first entry of $\partial L/\partial B$ is _____.

31. **[5 marks]** Consider one LoRA-adapted layer in a decoder-only language model. Use row vectors and the following shapes:

Quantity	Shape
x	1 × k
W_0	k × d
A	k × r
B	r × d
c	d × 1

The forward computation is:

$$u=xA,\quad h=xW_0+uB,\quad z=hc$$

W_0 and the readout vector c are fixed. A and B are trainable. The scalar loss L depends on x, W_0, A and B only through the scalar z. Let:

$$\delta=\frac{\partial L}{\partial z}$$

Define the upstream row gradient g as follows:

$$g=\frac{\partial L}{\partial h}$$

Fill in the five matrix expressions below. Each blank carries **1 mark**. You may use x, u, g, δ, W_0, A, B and c. Write gradients with respect to x, u and h as row vectors.

Part	Matrix expression
(a)	g = _____
(b)	$\partial L/\partial u$ = _____
(c)	$\partial L/\partial A$ = _____
(d)	$\partial L/\partial B$ = _____
(e)	$\partial L/\partial x$ = _____

Write the five final expressions in the corresponding blanks on the response sheet.